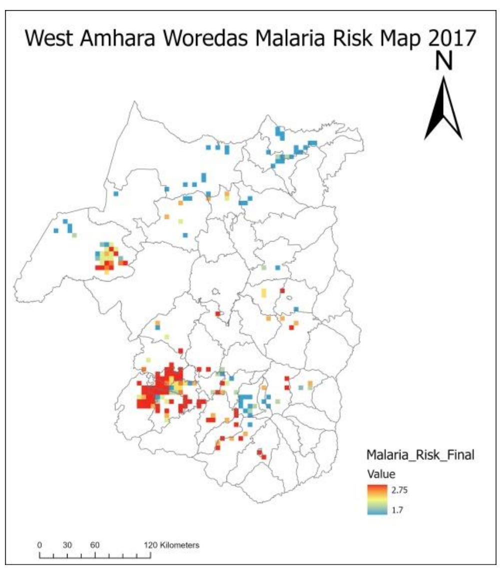
Bahir Dar University

Department of Geography and Environmental studies Masters in Geo-Information Science

Malaria risk Mapping and analysis of West Amhara, Woredas, Ethiopia.

Course Title: Computational and Geo-statistical Science (GeIs 632).



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1. INTRODUCTION

1.1 Background

Malaria remains one of the major public health problems in Ethiopia and continues to cause significant morbidity and mortality, particularly in lowland areas. The occurrence of malaria varies considerably from place to place due to differences in environmental and climatic conditions such as elevation, temperature, rainfall, vegetation, and surface moisture.

The West Amhara region is characterized by diverse topography and climate, ranging from hot lowlands to cool highlands. These variations create different levels of environmental suitability for mosquito breeding and malaria transmission. Understanding the spatial distribution of malaria and its associated environmental factors is important for identifying high-risk areas and supporting targeted malaria control interventions.

Geographic Information Systems and spatial statistical techniques provide useful tools for analyzing the distribution of malaria and its environmental determinants. Therefore, this study applies GIS, geostatistical methods, and spatial regression techniques to investigate the spatial distribution, environmental risk factors and determinants of malaria incidence in West Amhara, Ethiopia, using malaria data from 2017.

2 OBJECTIVES OF THE STUDY

General Objective

To investigate the spatial distribution, environmental suitability, and environmental determinants of malaria incidence in West Amhara, Ethiopia, using GIS, geostatistical techniques, and spatial regression methods.

Specific Objectives

1. To map the spatial distribution of malaria incidence in West Amhara using the Annual Parasite Index (API).
2. To identify areas environmentally suitable for malaria transmission using topographic and climatic factors.
3. To produce malaria risk maps by integrating environmental and climatic variables through weighted overlay analysis.
4. To determine whether malaria incidence exhibits significant spatial clustering and identify hotspot and cold spot areas.
5. To examine the relationships between malaria incidence and environmental variables using Ordinary Least Squares and Geographically Weighted Regression models.

3. STUDY AREA

The study was conducted in West Amhara, located in the northwestern part of the Amhara National Regional State, Ethiopia. The study area covers the administrative zones of North Gondar, South Gondar, West Gojjam, and East Gojjam and comprises approximately 121 Woredas and towns.

West Amhara is characterized by highly diverse topography, ranging from lowland plains along the Sudan border to high mountainous areas exceeding 2500 meters above sea level. The region exhibits considerable variations in climate, vegetation, rainfall, and temperature. The western lowlands are generally warm and receive favorable climatic conditions for malaria transmission, whereas the highland areas are relatively cool and less suitable for mosquito survival and parasite development.

The region experiences a seasonal rainfall pattern, with most precipitation occurring between June and September. Mean annual rainfall and temperature vary considerably across the study area because of differences in altitude and topography. The major land cover types include agricultural land, forest, shrub land, and grassland.

Malaria remains one of the major public health problems in West Amhara. Transmission intensity varies considerably from one Woreda to another due to environmental and climatic differences. The combination of low-lying areas, suitable temperatures, seasonal rainfall, and extensive agricultural activities creates favorable conditions for mosquito breeding and malaria transmission in several parts of the region.

The geographic diversity and varying malaria burden make West Amhara an appropriate study area for investigating the spatial distribution, environmental suitability, and determinants of malaria incidence using GIS and spatial statistical techniques.

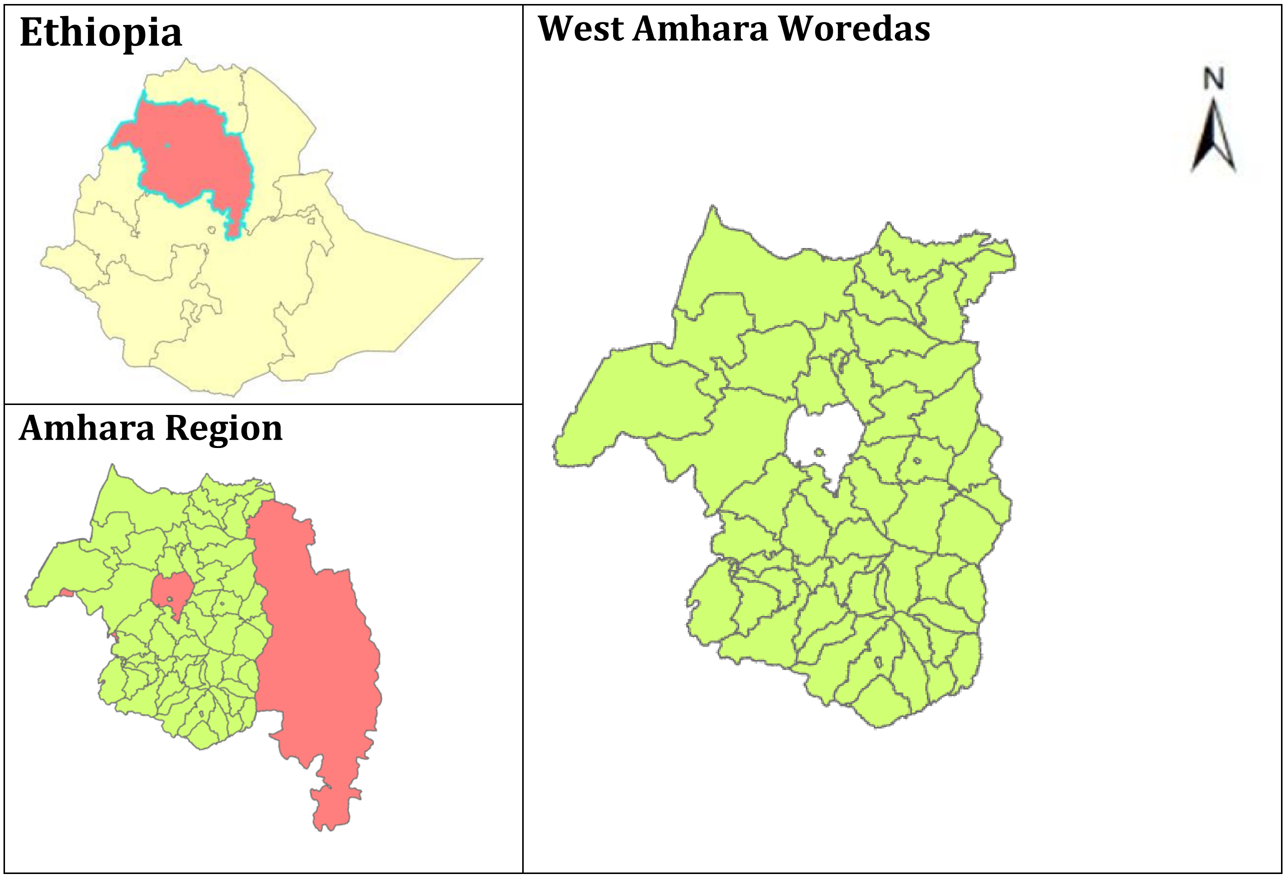


FIGURE 1:LOCATION MAP OF THE STUDY AREA IN WEST AMHARA, ETHIOPIA.

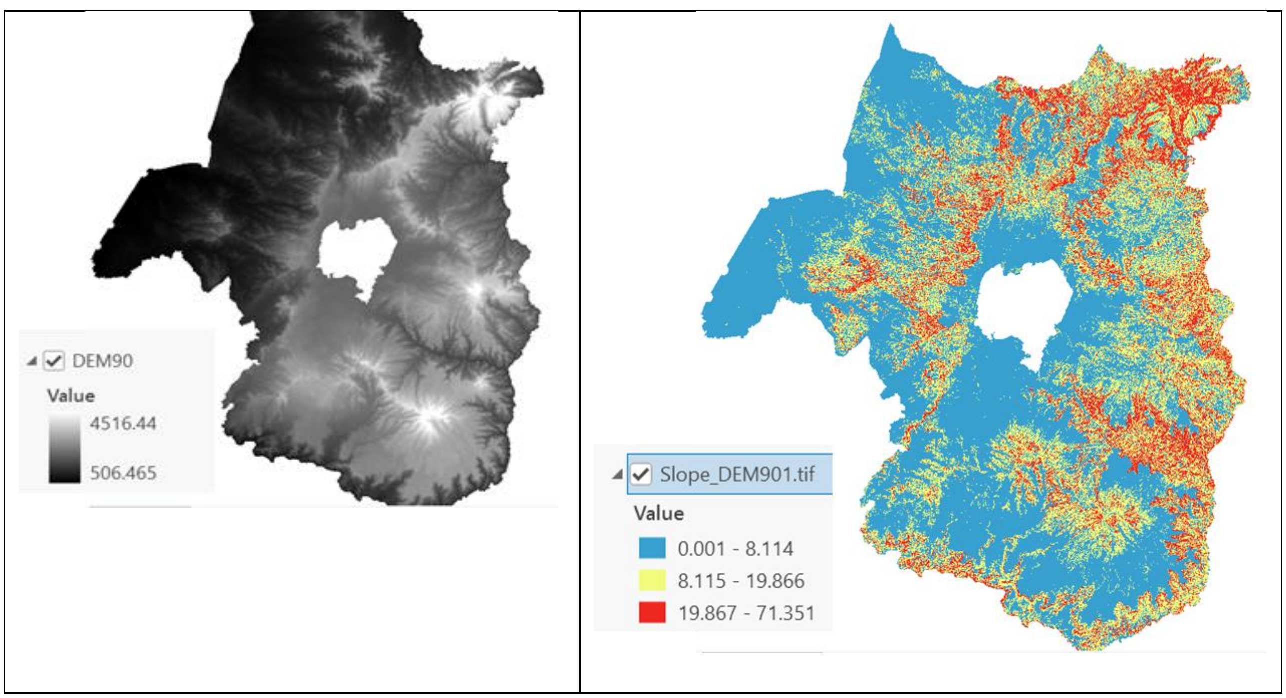


FIGURE 2:ELEVATION AND SLOPE OF STUDY AREA

TABLE 1:Geographic Characteristics Of The Study Area.

Characteristic	Description
Region	Amhara National Regional State
Study Area	West Amhara
Administrative Units	North Gondar, South Gondar, West Gojjam, and East Gojjam
Number of Woredas	56
Total Area	5225 Km^2
Total population	10,693,889
Elevation Range	506 to above 4500 m asl
Climate	Tropical highland and lowland climate
Main Economic Activity	Agriculture
Major Health Problem	Malaria

4. METHODOLOGY

Datasets

This study used health, environmental, climatic, and spatial datasets obtained from different sources to investigate the spatial distribution and environmental determinants of malaria in West Amhara. Malaria case and population data were used to calculate the Annual Parasite Index and analyze the spatial distribution of malaria incidence. Environmental and climatic datasets were used to develop malaria risk maps and examine the relationship between malaria incidence and environmental factors.

The environmental variables considered in this study included elevation, slope, topographic wetness index, vegetation cover represented by NDVI, rainfall, and mean temperature. These variables were selected because they directly or indirectly influence mosquito breeding, survival, and malaria transmission.

TABLE 2:Dasets Used In The Study

Category	Datasets
Health data	Malaria cases and population data
Environmental data	DEM, elevation, slope, TWI, and NDVI
Climate data	Rainfall station data, rainfall surface, and temperature

Spatial data	Administrative boundaries and spatial weights matrix
Derived products	MP_site, Clim_Risk, and Final Malaria Risk Map

4.1 MALARIA INCIDENCE DISTRIBUTION MAPPING

Malaria cases and population data for 2017 were used to calculate the Annual Parasite Index (API) for each Woreda and visualize patterns. The API values were mapped in ArcGIS Pro to examine the spatial distribution of malaria incidence and identify areas with high and low malaria burden.

4.2 ENVIRONMENTAL AND CLIMATIC RISK MAPPING

Environmental factors, including elevation, slope, topographic wetness index, and NDVI, were reclassified according to their suitability for malaria transmission and combined using weighted overlay analysis to produce the Mosquito Proliferation Suitability Map (MP_site). Similarly, rainfall and mean temperature were reclassified and combined to generate the Climate Risk Map (Clim_Risk). Finally, the two maps were integrated to produce the Final Malaria Risk Map.

WEIGHTED-OVERLAY

The Weighted-overlay Model and using Multi Criterion Evaluation (MCE) was applied to identify malaria risk areas. Identification of malaria risk areas is computed by using knowledge from existing literatures (Berhanu Berga , 2016),Instructor lab manual, and local knowledge to determine weight for the variables.

TABLE 3: Environmental Risk Factor Reclassification Ranges And Weights

Environmental Variable	Raw Values / Range	Reclassification Risk Level	Standardized Risk Score	Assigned Weight (%)
Altitude (Elevation)	< 1500 m	Moderate Risk	2	30%
	1500 - 2000 m	High Risk (Epidemic Fringe)	3	
	2000 - 2500 m	Low Risk	1	
	> 2500 m	Malaria-Free	0	
Slope & Euclidean Buffer	0 - 5° (with < 2000 m buffer)	High Risk	3	20%
	5 - 10°	Moderate Risk	2	
	> 10°	Low Risk	1	
NDVI & Flying Range	> 0.75 (with < 2000 m buffer)	High Risk	3	15%
	0.40 - 0.75	Moderate Risk	2	
	< 0.40	Low Risk	1	
Temperature	22 - 32°C (Optimal	High Risk	3	20%

	sporogony)			
	18 - 22°C	Moderate Risk	2	
	< 18°C (Sporogony inhibited)	Low Risk	1	
Rainfall	> 1200 mm	High Risk	3	15%
	1000 - 1300 mm	Moderate Risk	2	
	< 1000 mm	Low Risk	1	

MALARIA RISK MODEL BUILDER

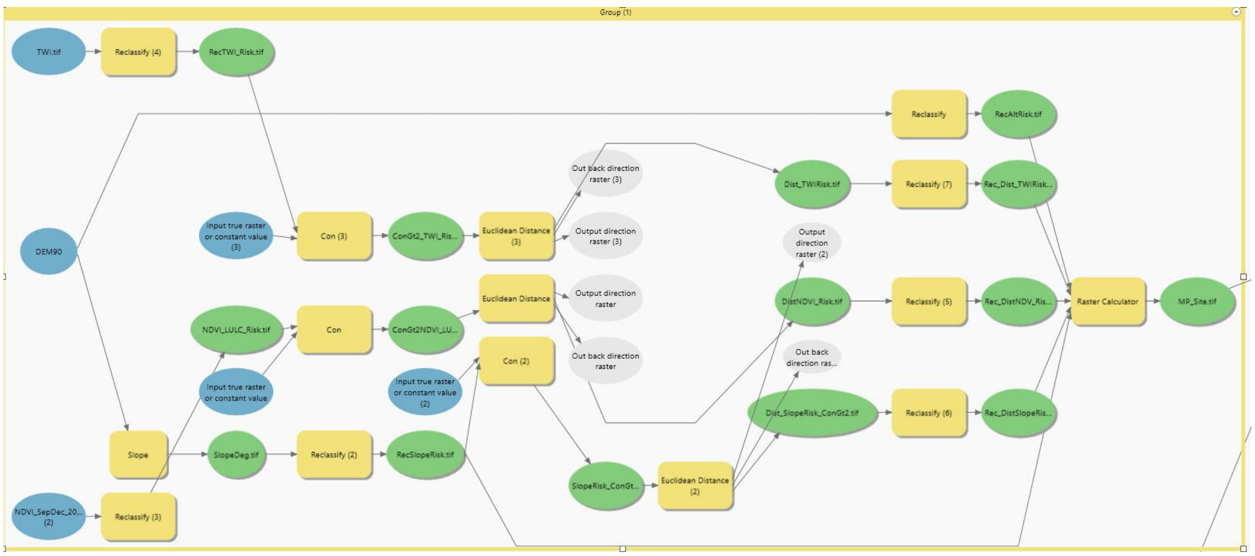


FIGURE 3: Model For Environmental And Climate Risk Maps(Partial View)

4.3 SPATIAL AUTOCORRELATION AND HOTSPOT ANALYSIS

Global Moran's I was used to determine whether malaria incidence exhibited significant spatial clustering across West Amhara. Local Moran's I was then applied to identify local clusters and spatial outliers, while Getis-Ord Gi star statistics were used to identify statistically significant hotspot and cold spot areas.

4.4 SPATIAL REGRESSION ANALYSIS

Ordinary Least Squares (OLS) and Geographically Weighted Regression (GWR) models were employed to examine the relationship between malaria incidence and environmental variables. OLS was used to investigate global relationships between malaria incidence and explanatory variables, whereas GWR was used to account for spatial non-stationarity and investigate how these relationships vary across geographic space. Model performance was evaluated using Adjusted R-squared, Akaike Information Criterion (AICc), and residual spatial autocorrelation diagnostics.

5. RESULTS AND DISCUSSION

5.1 SPATIAL DISTRIBUTION OF MALARIA INCIDENCE

The spatial distribution of malaria incidence in West Amhara showed considerable variation among Woredas during 2017. The Annual Parasite Index (API) ranged from approximately 4.30 to 23.76 cases per 1,000 population, indicating substantial spatial heterogeneity in malaria transmission.

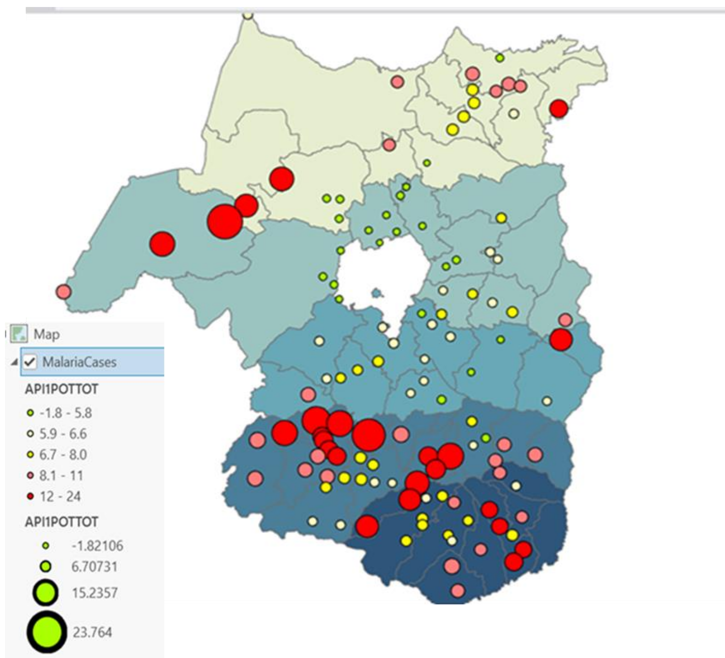


FIGURE 4:Spatial Distribution Of Malaria Incidence(Api)

The highest malaria incidence was observed in **Quara, Sekela, Fagta Lakoma, Bibugn, and Banja** Woredas, whereas relatively low API values were recorded in **Dembia, Wegera, Gondar, and Gondar Zuria.** The high-incidence areas are predominantly located in the western and southwestern parts of the study area, which are characterized by relatively warm temperatures and environmental conditions favorable for malaria transmission.

The observed spatial variability suggests that malaria incidence is not uniformly distributed across West Amhara and that certain areas experience a disproportionately higher malaria burden than others. These findings indicate the need for geographically targeted malaria prevention and control interventions.

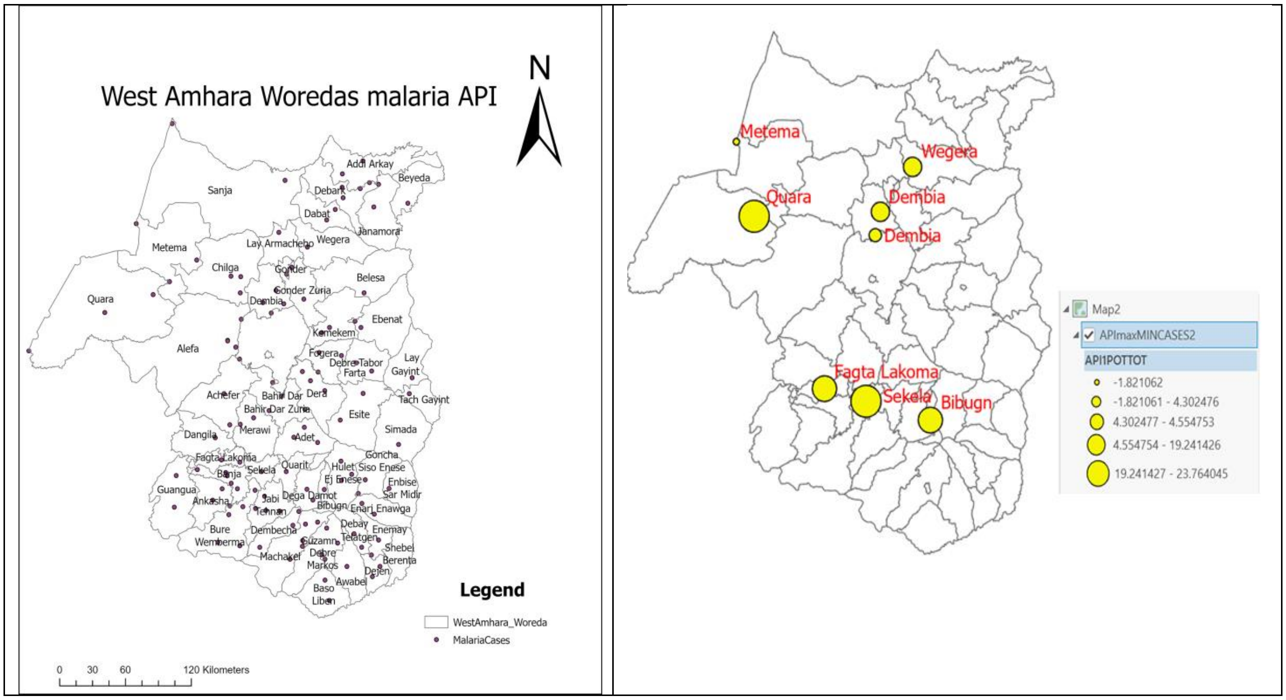


FIGURE 5:Distribution Of Highest And Lowest Malaria Incidence(Api)

Analyses show substantial spatial heterogeneity in malaria incidence, with the highest Parasite Index (API) values observed in **Quara,Sekela,Fagta Lakoma,Bibugn and Banja Woredas.**

TABLE 4:Comparison Of Highest And Lowest Malaria Incidence(API)

Highest API	API	Lowest API	API
Quara	23.76	Dembia	4.30
Sekela	22.22	Wegera	4.47
Fagta Lakoma	19.24	Gondar	5.00
Bibugn	17.87	Gondar Zuria	4.65
Banja	17.21	Dembia	4.55

5.2 ENVIRONMENTAL AND CLIMATIC RISK MAPPING

Environmental and climatic factors were integrated using weighted overlay analysis to identify areas that are suitable for malaria transmission in West Amhara. The environmental suitability map (MP_site) was generated from elevation, slope, topographic wetness index, and NDVI, while the climate risk map (Clim_Risk) was produced using rainfall and mean temperature both interpolated using the Co-Kriging method. CoKriging method was used after comparing the performance and RMSE of the three methods.

- IDW = 95.94
- Kriging = 76.2
- CoKriging = 62.9

The main parameters that influence RMSE difference were the Power parameter, Transformation(Log) and the use of Optimization tool.

Based on these weights given (Table 2), the final malaria risk map was produced using Weighted Overlay Model (Figure 6).

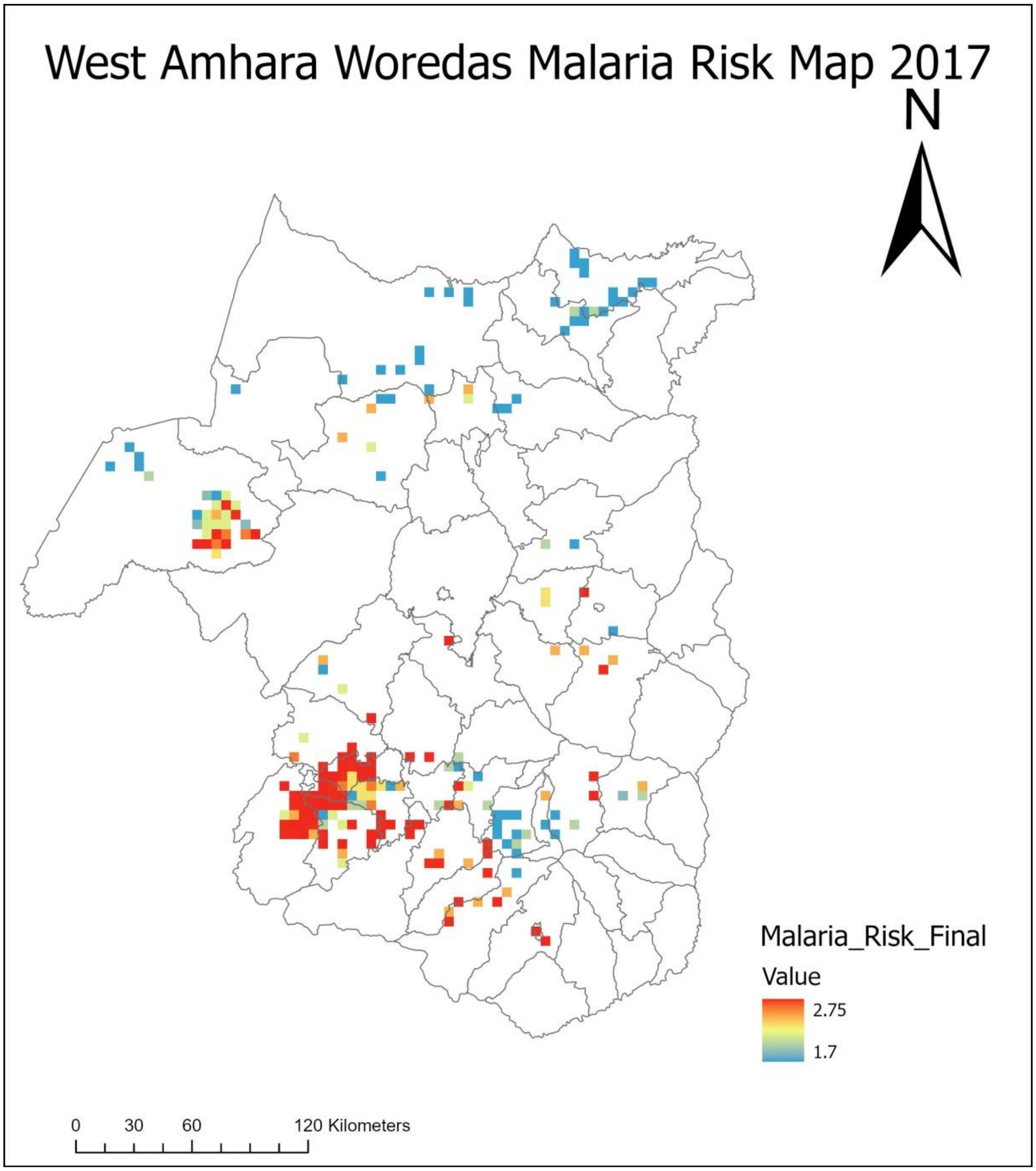


FIGURE 6:Final Malaria Risk Map Of West Amhara.

Table below shows the top five Woredas with the highest modeled malaria risk after performing Zonal statistics.

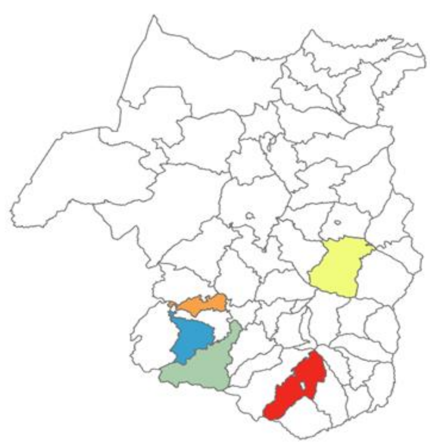
TABLE 5:Top Five Woredas With The Highest Malaria Risk

Woreda	Mean Risk
Esite	2.65
Guzamn	2.65
Fagta Lakoma	2.63
Bure Wemberma	2.60
Ankasha	2.58

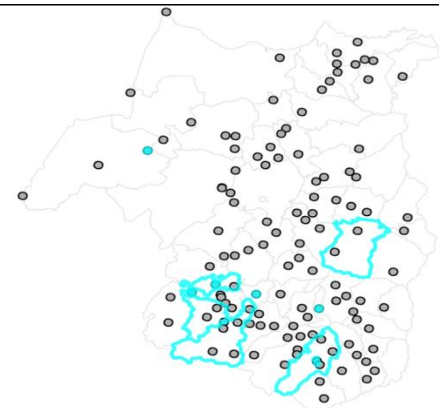
COMPARISON BETWEEN OBSERVED VS MAPPED

As shown in the maps below, the final malaria risk map showed considerable spatial variation across the study area. Areas with high and very high malaria risk were mainly concentrated in the western and southwestern parts of West Amhara, whereas the central and northern highlands were generally characterized by low malaria risk. The Woredas with the highest modeled malaria risk included **Esite, Guzamn, Fagta Lakoma, Bure Wemberma, and Ankasha**. These areas possess favorable environmental and climatic conditions that support mosquito breeding and malaria transmission.

In contrast, the spatial pattern of the final risk map generally agrees with the distribution of observed malaria incidence, particularly in the western lowlands where both high malaria risk and high API values were recorded. However, the agreement with specific Woredas identified was not perfect, suggesting that environmental suitability alone cannot fully explain malaria occurrence and that additional factors such as human activities, interventions, and socioeconomic conditions may also influence malaria transmission.



Identified Top 5 High Malaria risk Woredas



Identified Top 5 High Malaria risk Woredas and top 5 API observed points selected and overlaid

FIGURE 7:Comparison Between Observed Vs Mapped

5.3 SPATIAL AUTOCORRELATION AND HOTSPOT ANALYSIS

GLOBAL MORAN'S I

Global Moran's I analysis revealed a Moran's I value of 0.28 with a statistically significant p-value, indicating that malaria incidence in West Amhara is spatially clustered rather than randomly distributed. These results suggest that neighboring Woredas tend to exhibit similar malaria incidence patterns.

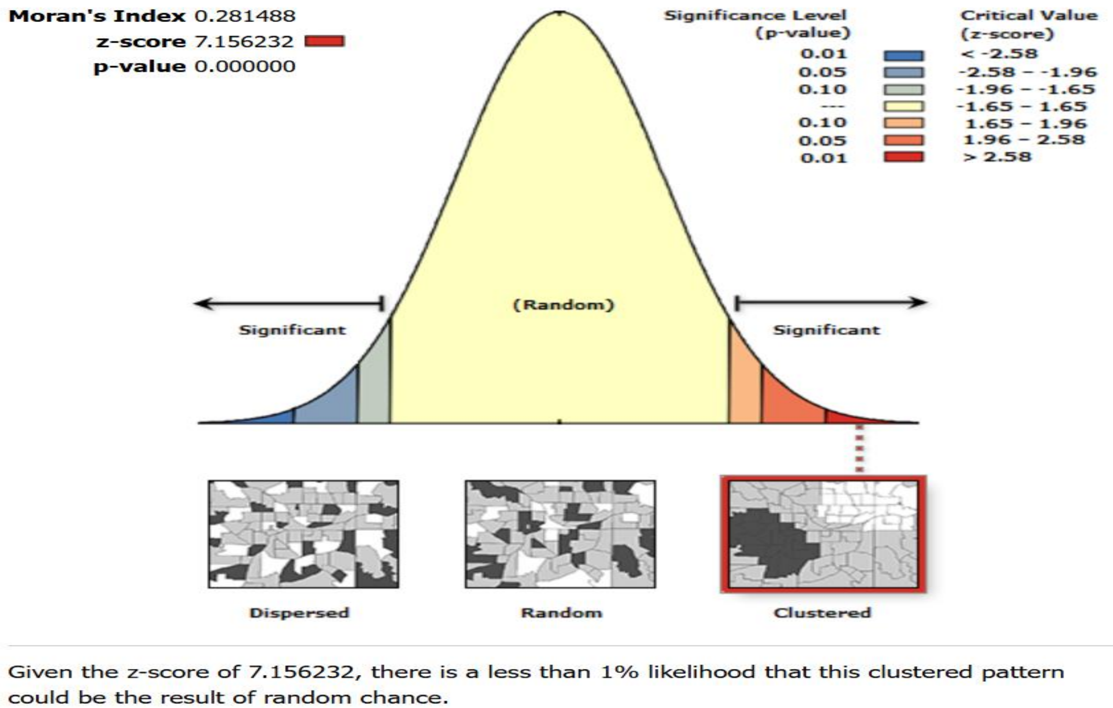


FIGURE 8:Global Moran's I Result For Malaria Incidence.

- Global spatial autocorrelation analysis using Moran's I (0.28, p = 0.00) confirmed statistically significant clustering of malaria cases across the study area.
- Malaria incidence exhibits statistically significant positive spatial autocorrelation and is clustered rather than randomly distributed.

LOCAL MORAN'S I AND GETIS-ORD GI

To identify the specific locations of these clusters, Local Moran's I and Getis-Ord Gi star analyses were performed. Both Local Moran's I and Getis-Ord Gi* analyses identified significant hotspots and cold spots Woredas, indicating strong localized spatial dependence.

Local Moran's I identifies local clusters and spatial outliers by comparing each Woreda with its neighboring Woredas. The method classifies areas into high-high, low-low, high-low, and low-high categories. In contrast, Getis-Ord Gi star analysis focuses on identifying statistically significant concentrations of high values (hotspots) and low values (cold spots). Therefore, Local Moran's I emphasizes local similarity and outlier detection, whereas Getis-Ord Gi star emphasizes the intensity and concentration of malaria incidence.

This can be seen better by different ways such as displaying the HH, HL,LL,LH of Local Moran's I values ;and the point based clustering of Getis-Ord Gi* value; or looking the heat map of both as shown below.

FIGURE 9:LOCAL MORAN'S I CLUSTER AND SPATIAL OUTLIER MAP.

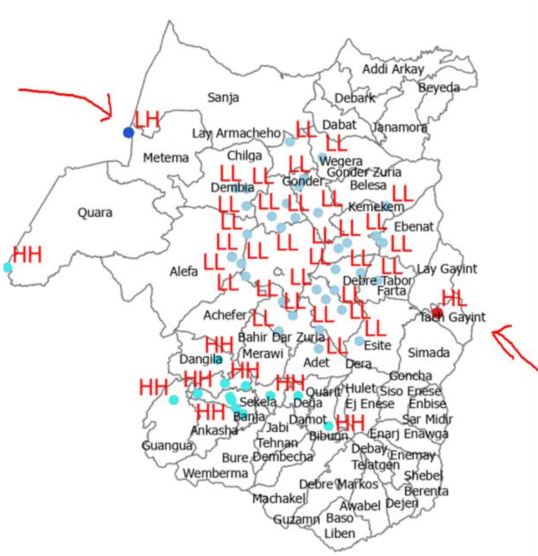


FIGURE 10:GETIS_ORD_HOTSPOTS

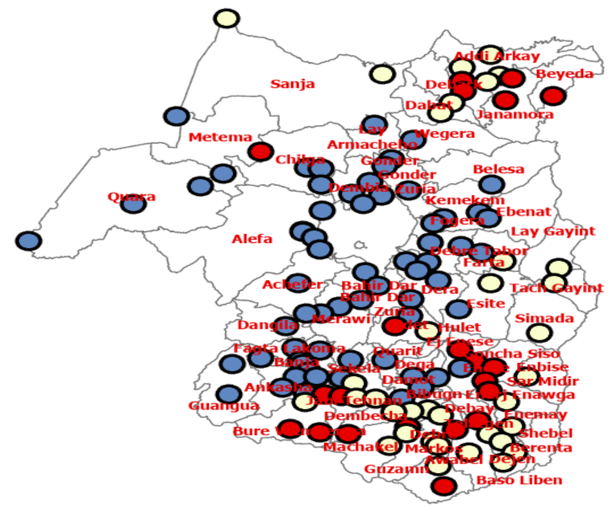


FIGURE 11:HEAT MAP OF LOCAL MORAN'S I

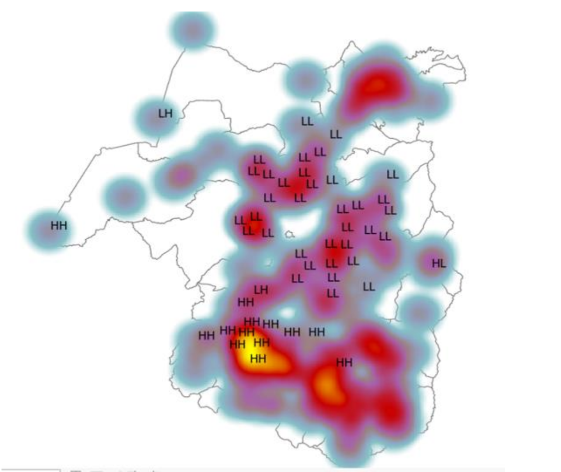
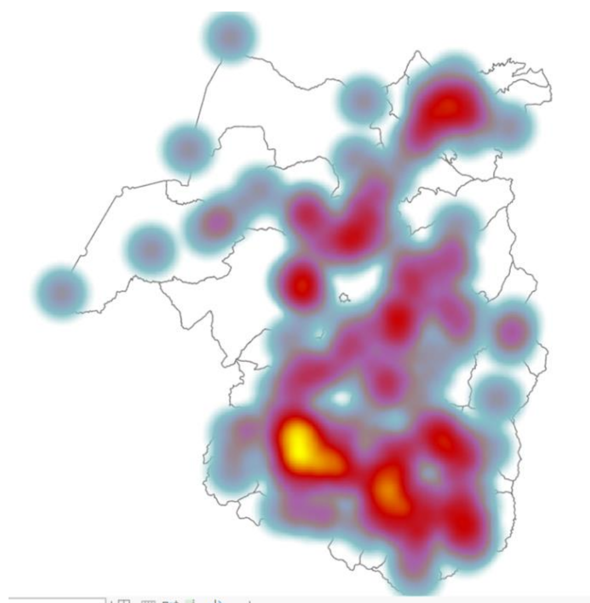


FIGURE 12:HEAT MAP OF GI* VALUES



Local Moran's I identified significant **high-high(HH)** clusters in **Fagta Lakoma, Banja, Sekela, Guangua, and Dega Damot**, indicating areas with high malaria incidence surrounded by neighboring areas that also experience high malaria incidence. Significant **low-low(LL)** clusters were identified in Gondar, Dembia, Gondar Zuria, Chilga, Fogera, and Bahir Dar Zuria, representing areas with consistently low malaria incidence.

The Getis-Ord Gi star hotspot analysis further confirmed the existence of significant malaria hotspots in **Fagta Lakoma, Banja, Sekela, Guangua, and Dangila**. Significant cold spots were observed in **Gondar, Dembia, Gondar Zuria, and Chilga.** The concentration of hotspots in the western and southwestern parts of the study area indicates that malaria transmission is geographically concentrated in specific areas rather than being uniformly distributed across West Amhara.

Outliers

- Some Woredas shown by the arrows exhibited unique spatial behavior compared to surrounding areas. For example, Metema was identified as a low-high spatial outlier in the Local Moran's I analysis, indicating that the Woreda showed relatively low malaria incidence while being surrounded by neighboring areas with relatively higher malaria incidence. Such spatial outliers may reflect localized environmental conditions, intervention effectiveness, reporting differences, or population-related factors.
- Quara, on the other hand, exhibited hotspot characteristics in the Getis-Ord Gi star analysis and high-high clustering patterns in the Local Moran's I analysis, indicating that the Woreda and its neighboring areas consistently experience elevated malaria incidence.

The identified hotspot areas should therefore be prioritized for malaria surveillance, vector control, and resource allocation. The clustering of malaria incidence also suggests that environmental and geographic factors play an important role in shaping malaria transmission patterns in the study area.

5.4 SPATIAL REGRESSION ANALYSIS

ORDINARY LEAST SQUARES

Ordinary Least Squares and Geographically Weighted Regression models were employed to investigate the relationship between malaria incidence and environmental variables in West Amhara. The explanatory variables included elevation, slope, topographic wetness index, NDVI, rainfall, and mean temperature. Model performance and interpretation were evaluated using diagnostic indicators including Adjusted R-squared, Akaike Information Criterion (AICc), residual distribution, and spatial autocorrelation measures.

The Ordinary Least Squares model indicated that the selected environmental variables are associated with malaria incidence; however, the model exhibited limited explanatory power and assumed that the relationships between malaria and environmental factors remain constant throughout the study area. This is indicated in the result table below by KPIs.

Among the explanatory variables, NDVI showed statistically significant association with malaria incidence ($p < 0.05$), while the remaining variables were not statistically significant in the OLS model. All variables showed acceptable VIF values below 5, indicating low multicollinearity.

Summary of OLS Results - Model Variables

Variable	Coefficient [a]	StdError	t-Statistic	Probability [b]	Robust_SE	Robust_t	Robust_Pr [b]	VIF [c]
Intercept	-2.031230	2.618038	-0.775859	0.439432	3.040612	-0.668033	0.505461	--------
MEAN_NDVI	21.290003	4.374647	4.866679	0.000004*	4.467347	4.765692	0.000007*	1.022667
MTWI_MEAN	-0.134287	0.405979	-0.330774	0.741430	0.328100	-0.409287	0.683106	1.065480
DEM_MEAN	-0.000079	0.000672	-0.117232	0.906876	0.001028	-0.076658	0.939022	1.178032
SLOPEMEAN	0.043638	0.071953	0.606478	0.545405	0.065045	0.670894	0.503643	1.150980
TMEAN	-0.006610	0.034942	-0.189176	0.850290	0.029881	-0.221217	0.825320	1.084538
RFMEANCOGR	-0.000389	0.000527	-0.737493	0.462333	0.000461	-0.844438	0.400185	1.067300

FIGURE 13:Ordinary Least Squares Model Results.

GEOGRAPHICALLY WEIGHTED REGRESSION

To account for spatial variation in these relationships, a Geographically Weighted Regression model was developed. The GWR model showed improved performance compared to the global OLS model, with an adjusted R-squared value of 0.515 and a lower Akaike Information Criterion value. This indicates that approximately 51 percent of the spatial variation in malaria incidence can be explained by the environmental variables included in the model.

TABLE 6: OLS VS. GWR Model Fitness Performance Comparison

Model Evaluation Diagnostic Metric	Global Ordinary Least Squares (OLS)	Local Geographically Weighted Regression (GWR)
Model Fitness (Adjusted R^2)	**0.17**	**0.515**

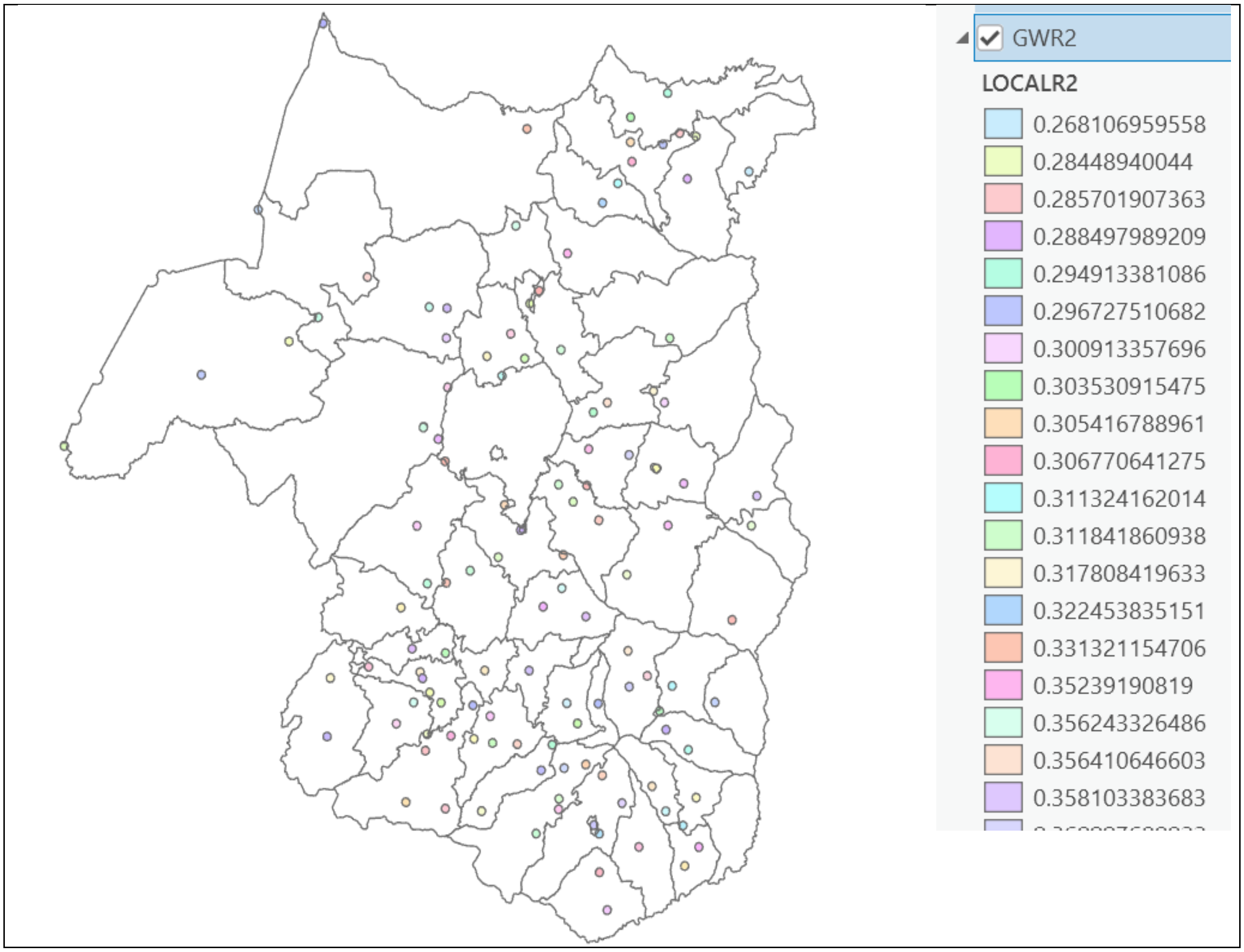


FIGURE 14:Geographically Weighted Regression Local R-Squared Map.

The results suggest that environmental factors play an important role in shaping malaria distribution in West Amhara, but they do not fully explain the observed malaria patterns. The remaining variation may be associated with other factors that were not included in the study, such as socioeconomic conditions, population movement, healthcare access, malaria interventions, and human behavior.

The improved performance of the GWR model further indicates that the relationships between malaria incidence and environmental variables vary across geographic space. Therefore, the influence of environmental factors on malaria transmission is not uniform throughout West Amhara, and geographically targeted malaria interventions may be more effective than uniform regional approaches.

EXPLORATORY REGRESSION(ER)

Explanatory Regression allows identify suitable combinations of environmental variables for modeling malaria incidence. Multiple candidate models were tested automatically using adjusted R-squared, AICc, VIF, and spatial autocorrelation diagnostics.

```
Choose 1 of 6 Summary
    Highest Adjusted R-Squared Results
AdjR2   AICc    JB  K(BP)  VIF   SA    Model
 0.17 660.25 0.00  0.41 1.00 0.00  +MEAN_NDVI***
 0.00 682.44 0.00  0.08 1.00 0.00  -RFMEANCOGR
-0.00 683.25 0.00  0.82 1.00 0.00  +SLOPEMEAN
       Passing Models
AdjR2 AICc JB K(BP) VIF SA   Model
******************************************************************************
Choose 2 of 6 Summary
           Highest Adjusted R-Squared Results
AdjR2   AICc    JB  K(BP)  VIF   SA    Model
 0.17 661.57 0.00  0.31 1.01 0.00  +MEAN_NDVI***   -RFMEANCOGR
 0.17 661.87 0.00  0.51 1.00 0.00  +MEAN_NDVI***   +SLOPEMEAN
 0.17 662.08 0.00  0.56 1.00 0.00  +MEAN_NDVI***   -MTWI_MEAN
       Passing Models
AdjR2 AICc JB K(BP) VIF SA   Model
******************************************************************************
Choose 3 of 6 Summary
                 Highest Adjusted R-Squared Results
AdjR2   AICc    JB  K(BP)  VIF   SA    Model
 0.17 663.27 0.00  0.40 1.01 0.00  +MEAN_NDVI***   +SLOPEMEAN   -RFMEANCOGR
 0.16 663.58 0.00  0.44 1.04 0.00  +MEAN_NDVI***   -MTWI_MEAN   -RFMEANCOGR
 0.16 663.63 0.00  0.11 1.03 0.00  +MEAN_NDVI***   -TMEAN   -RFMEANCOGR
       Passing Models
AdjR2 AICc JB K(BP) VIF SA   Model
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FIGURE 15:Explanatory Regression Report Summary

- Results showed that NDVI consistently appeared in the best-performing models, indicating strong explanatory contribution to malaria distribution.
- No global model achieved high explanatory power. However, NDVI was the strongest globally significant predictor.
- Variables including rainfall, slope, and TWI also appeared in several acceptable models and were retained for further regression analysis.

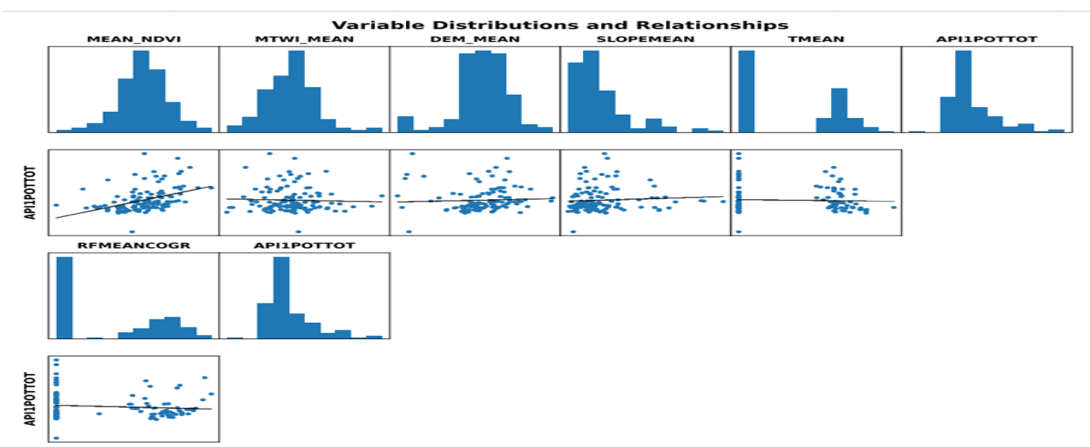

FIGURE 16:Explanatory Regression Variable Distribution And Relationship

6. CONCLUSION AND RECOMMENDATIONS

6.1 CONCLUSION

This study investigated the spatial distribution, environmental suitability, and environmental determinants of malaria incidence in West Amhara, Ethiopia, using GIS, geostatistical techniques, and spatial regression methods.

The results revealed considerable spatial heterogeneity in malaria incidence across the study area. High malaria incidence was mainly concentrated in the western and southwestern parts of West Amhara, particularly in Quara, Sekela, Fagta Lakoma, Bibugn, and Banja.

Environmental and climatic risk mapping identified several Woredas as highly suitable for malaria transmission, including Esite, Guzamn, Fagta Lakoma, Bure Wemberma, and Ankasha. Although the modeled risk map generally agreed with the observed malaria distribution, the two patterns were not identical, indicating that environmental suitability alone cannot fully explain malaria occurrence.

Global Moran's I and local spatial analyses confirmed that malaria incidence is significantly clustered rather than randomly distributed. Significant hotspots were identified in Fagta Lakoma, Banja, Sekela, Guangua, and Dangila, while cold spots were observed in Gondar, Dembia, Gondar Zuria, and Chilga.

Spatial regression analysis demonstrated that environmental variables explain part of the observed malaria pattern in West Amhara. The Geographically Weighted Regression model performed better than the global model and indicated that approximately 51 percent of the spatial variation in malaria incidence can be explained by the selected environmental variables. This finding suggests that other factors, including socioeconomic conditions, human activities, and malaria interventions, also contribute to malaria transmission.

Overall, the study demonstrates that malaria transmission in West Amhara exhibits strong spatial heterogeneity and that integrating GIS, spatial statistics, and environmental analysis provides valuable information for identifying high-risk areas and supporting geographically targeted malaria control strategies.

6.2 RECOMMENDATIONS

1. Malaria prevention and control activities should prioritize hotspot areas, particularly Fagta Lakoma, Banja, Sekela, Guangua, and Dangila.

2. Health authorities should allocate resources and intervention programs according to the spatial distribution of malaria risk rather than applying uniform interventions across all Woredas.

3. Future studies should incorporate additional variables such as population density, irrigation, land use, socioeconomic conditions, healthcare accessibility, and malaria intervention data to improve model performance.

4. Multi-year malaria and environmental data should be used in future studies to better understand temporal variations in malaria transmission.

5. The final malaria risk map should be periodically updated using recent environmental and epidemiological data to support evidence-based malaria surveillance and decision-making.

References

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